Tuni2025 GNSS Spoofing Datasets

Scenario: GPS Static No Multipath True Position – Clear-Sky (C-7)

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1. Overview

This dataset is part of the Tuni2025 GNSS measurement campaign conducted at Tampere University, Finland. It represents a **clean GPS-only environment** with **true-position signals**, recorded under static conditions with **no spoofers**.

The dataset includes:

- Raw I/Q GNSS signal measurement file
- This README file
  (No scenario configuration file is provided)
- default_param_GPSL1_Chapter2.txt

  FGI GSRX configuration file for processing the GPS measurements

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2. Measurement and Scenario Details

Parameter	Description
Measurement Date	30-01-2025
Measurement Time	18:30 (local time)
Scenario Type	GPS Static, True Position, Clear-Sky
Spoofers Present	None
Signals Present	GPS L1 signals only
Multipath Condition	Not Present
Measurement Device	USRP-2945R SDR front-end
Receiver Gain	20 dB (for real signals)
File Format	Raw I/Q data (interleaved 32-bit float, .bin)
Sampling Frequency	50 MSps

Parameter	Description
Scenario File	Not included

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3. File(s) Included

- CLearsky signal C-7.bin
  (Approx. 30 GB)
  Raw I/Q data containing GPS L1 signals, under clear-sky and spoofer-free conditions.
- This README file
  (Containing detailed description of the scenario)
- default_param_GPSL1_Chapter2.txt

  FGI GSRX configuration file for processing the GPS measurements

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4. Usage Notes

- Serves as a benchmark for evaluating spoofing detection algorithms in multipath environments.
- Can be used as a control case in experiments distinguishing between spoofing and natural multipath effects.
- Ensure sufficient storage and memory when working with high-sample-rate data.

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5. Citation

If you use this dataset, please cite it.

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6. License

CC BY 4.0 — This dataset is freely reusable with appropriate credit.